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Young Children's Experience in Unplugged Activities About Computational Thinking: From an Embodied Cognition Perspective

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Abstract
Researchers are increasingly calling for more computational thinking (CT) teaching tools and activities designed for young children. Considering young children's need to draw on their bodily experiences to learn abstract concepts, this study applied the embodied cognition perspective to design an unplugged (non-computer-based) toolkit with activities that foster CT. This study aimed to explore young children's experiences in unplugged activities. Specifically, children's perceptions about the activities were investigated, and their embodied interactions during the activities were analyzed to reveal how CT emerges as an embodied phenomenon. Results indicate that children had positive perceptions about the activities. In addition, young children's algorithmic thinking and debugging emerged most frequently. Furthermore, this study found that CT extended to both the unplugged toolkit and children's bodies and emerged in perception-action loops. This research provides insight into the instructional design of young children's CT and helps researchers understand how young children develop CT from an embodied cognition perspective.

Keywords Computational thinking · Young children · Unplugged activities · Embodied interaction · Embodied cognition

Introduction

In the technology-driven world, computational thinking (CT) is expected to become a basic competence that everyone should master, like reading, writing, or arithmetic (Wing, 2006). Researchers have reached a consensus that students at all educational levels, from pre-primary to higher education, should learn CT as a fundamental, twenty-first century literacy (Grover & Pea, 2013; Shute et al., 2017). Although researchers emphasize that it is necessary to teach CT from early childhood, there are few studies focusing on the development of young children's CT (Hsu et al., 2018; Lee et al., 2022). Moreover, researchers are increasingly calling for more CT teaching tools and activities to be designed for young children (Manches & Plowman, 2017; Relkin et al., 2021).

Young children do not have mature literacy, arithmetic, and abstract reasoning skills, and they cannot fully master abstract concepts and complex logic without physical representations (Beilin, 1971). Thus, it is a great challenge for educators to develop tools and activities that promote young children's CT. Some researchers claim that embodied cognition may offer insight into CT education for young children (Huang & Looi, 2021; Manches et al., 2020). Embodied cognition refers to a group of hypotheses in cognitive science, most of which advocate that humans are able to acquire conceptual knowledge from sensorimotor experience gained from their bodily interactions with their environment (Borghi et al., 2011). An embodied cognition perspective may help young children learn because it emphasizes the way learners draw on their bodily experience to understand abstract concepts (Kopcha et al., 2021) and stresses that cognition is grounded in the physical environment (Alibali & Nathan, 2012). As such, this perspective accommodates young children's reliance on their physical environment.

The embodied cognition perspective is reflected in the teaching of CT. The programming language LOGO uses the image of a "turtle" as a virtual body agent (Bull et al.,

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2020). Similarly, some graphical programming languages, such as Scratch, a common tool for teaching CT to children today, use virtual characters (del Olmo-Muñoz et al., 2020). Children need to program to control these virtual objects' movements. Although students do not actually use their bodies in the learning process, they need to use past physical experiences and internal resources to solve problems. In addition to these plugged activities which provide children with computer programming exercises (Brackmann et al., 2017), some unplugged activities also reflect the embodied cognition perspective. In unplugged activities, there is no use of computers; learners learn from kinesthetic experiences (Webb et al., 2017). For example, students may play the role of a robot and complete a series of actions under specific commands from their teacher to learn CT concepts, such as sequence (Saxena et al., 2020). Researchers offer empirical evidence that a greater degree of bodily engagement could provide concrete experiences and support the perceptual experiences of young children (Sung et al., 2017; Johnson-Glenberg et al., 2014). This evidence highlights the advantages of using unplugged activities in teaching CT to young children compared to plugged activities like graphical programming.

However, whether using plugged or unplugged tools, little is known about young children's dynamic learning process of CT, as most research has focused on learning outcomes (Brackmann et al., 2017; Città et al., 2019; del Olmo-Muñoz et al., 2020; Hsu & Liang, 2021). Meanwhile, scholars assert that future works should explore how children might embody CT (Brennan & Resnick, 2012; Grover & Pea, 2013; Yasar, 2018). To address these research gaps, this study attempts to design an unplugged toolkit along with activities based on embodied cognition perspectives to help young children learn CT. Additionally, this study explores children's experiences in unplugged activities by analyzing their embodied interaction with their learning environment, revealing how children experience CT as an embodied phenomenon during these activities.

Literature Review

Teaching Computational Thinking to Young Children

Originally, Computational Thinking (CT) was defined as thinking that involves solving problems, designing systems, and understanding human behavior by drawing on concepts fundamental to computer science (Wing, 2006). Currently, researchers generally agree that computational concepts and computational practices are the core components of CT (Brennan & Resnick, 2012; Barr & Stephenson, 2011). Computational concepts refer to the means of solving computational problems, such as sequences, loops, conditionals, and events (Brennan & Resnick, 2012; Kotsopoulos et al., 2022), while computational practices refer to the methods of solving computational problems, such as abstraction, decomposition, and algorithmic thinking (Brennan & Resnick, 2012; Lee et al., 2022; Selby, 2013).

Many countries have emphasized the significance of teaching CT to K12 children (Hsu et al., 2018). However, little attention is paid to children three to eight years of age in existing research (Hsu et al., 2018; Lee et al., 2022), and scholars have called for more research working on tools and pedagogy to help educators teach CT to young children (Relkin et al., 2021). Taking into consideration young children's capacity for symbolic but not logical thought, Sigelman and Rider (2021) proposed that young children should be provided opportunities for interaction with their physical environment. Having a similar perspective, Asher (1977) emphasized the essence of learning through physical actions. Supported by this view, unplugged activities could be an effective teaching method to foster young children's CT. In such activities, children learn from kinesthetic experiences where they can focus on understanding concepts without being distracted by mastering tools and techniques like programming (Webb et al., 2017). Furthermore, researchers have noted that unplugged activities have unique advantages for novice teachers and children; they could lessen concerns related to technology access and the appropriateness of technology use in the early years (Kotsopoulos et al., 2022).

Some researchers have provided evidence that teaching CT with unplugged activities could effectively enhance young children's CT. There are two main forms of these unplugged activities. One is the physical enactment of predefined scenarios or sequences that contain specific cues for movement (Fadjo, 2012). For example, Saxena et al. (2020) designed an unplugged activity named "Tic-Tac-Toe" in which students act as a robot and follow teachers' directional commands to learn sequence. By observing the degree of completion of students' tasks, teachers scored students' CT. Similarly, Città et al. (2019) used "Robot-Tino Walk," a role-playing game in which some students act as robots while others design commands, to teach algorithm and debugging. The other main form of unplugged CT activity involves the learner physically controlling and manipulating the movement of an external surrogate, such as a virtual character or physical object (Fadjo, 2012). Hsu and Liang (2021) designed an unplugged board game in which students move tangible characters to corresponding positions through instruction cards about CT concepts. Furthermore, Chen and Chi (2020) designed the board game "Code Ocean" in which students use cards related to CT concepts to seize their opponents' treasure. However, both forms of activities have limitations. Role-playing games involve limited CT concepts and practices, and neither activity can provide feedback to learners. Furthermore, previous studies have demonstrated

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the positive effect of unplugged activities, but most focus on the learning outcome rather than the learning process.

Embodied Cognition and Computational Thinking

Many unplugged activities have been designed to engage children with body-based analogies of computational concepts, and their effects have been confirmed to some extent. Cortina (2015) argued that these activities are effective because children are physically part of the solution to a problem, learning by observing and experiencing. Manches et al. (2020) noted that no clear theoretical framework was provided for why certain body-based experiences might develop CT. In response to this issue, researchers claimed that embodied cognition could offer a particular perspective (Huang & Looi, 2021; Manches et al., 2020). Embodied cognition is an umbrella term capturing a class of theories within cognitive science, many of which support the idea that cognition is grounded in the sensorimotor activity of our bodies (Barsalou, 2008; Wilson, 2002; Abrahamson & Lindgren, 2014). Compared to the mainstream cognition theories before the 1980s, which assumed that the mind is an abstract information processing system separate from the brain's modal systems for perception and action, embodied cognition assumed that human cognitive processes are rooted in perceptual and physical interactions of the human body with the world (Barsalou, 2008; Manches et al., 2020). Wilson (2002) combed the literature on embodied cognition and concluded six prominent claims, which clarified the function of cognition and the relationship between cognition, environment, and body. Five of these refer to how humans offload cognitive work to the environment and body to reduce cognitive workload. The remaining claim emphasizes that the function of the mind is to guide action.

These insights regarding embodied cognition inspired educational researchers on how to teach and learn (Skulmowski & Rey, 2018). Based on the embodied cognition perspective, scholars further highlighted that the learning of abstract concepts is supported by bodily activities interacting in a perceived physical world (Barsalou, 2008; Glenberg, 2010). Considering that kinesthetic interactions grounded in concepts lead to conscious thinking, researchers regarded an embodied approach as a valuable way to teach abstract and symbolic concepts or literacies, such as programming languages and mathematics, to novice learners (Sung et al., 2017). As for how to effectively apply embodied cognition in learning environments, Abrahamson and Lindgren (2014) proposed specific guidelines regarding three aspects: activities, materials, and facilitation. Researchers have noted that it is critical to design a learning environment that naturally leads learners to take action in ways that simulate the essential concepts or mechanisms of the learning objective (Abrahamson & Lindgren, 2014).

However, few scholars have tried to introduce embodied cognition into the teaching of CT. Sung et al. (2017) conducted research about teaching math and CT to kindergarten and first grade children, finding that full-embody activities combined with the practice of computational perspective-taking greatly improved mathematics understanding and programming skills. Clarke-Midura et al. (2021) drew from the embodied cognition framework in the design of curricular tasks to teach kindergarten kids how to program using robots and studied the reference frames used by young children in toy-based coding. Both studies used embodied methods to promote young children's CT, but they ignored how CT is reflected in children's embodied interaction. Kopcha et al. (2021) put forward an analysis framework to explore how children's embodied interaction behaviors embody CT. Using this framework, they analyzed recordings of two fifth-grade learners' performances in educational robot activities. Furthermore, Kopcha et al. (2021) called on future researchers to use this framework to analyze different learning contexts. In general, scarce research explores how young children embody CT via unplugged activities designed from embodied cognition perspectives.

Therefore, this study designed an unplugged toolkit named CodeWay with supporting activities, following the guidelines proposed by Abrahamson & Lindgren (2014), to help young children learn CT. Furthermore, this study focused on young children's embodied experience in these activities. Specially, the research questions are stated as follows:

RQ1. Do the students' perceptions of unplugged activities improve after engaging with them?
RQ2. What's the frequency of each students' CT emerging during unplugged activities?
RQ3. How do young children experience CT as an embodied phenomenon during unplugged activities?

Method

Participants

This study recruited 28 Chinese children (ages 6–8) to participate in the experiment. Their parents knew the content of this experiment and agreed to their participation. Parents also completed a short demographic questionnaire (see Table 1). Furthermore, all of the children's families lived in Beijing, China, belonged to the middle class, and the children had never learned programming or taken other courses related to CT. To know participants' prior level of CT, a test was conducted within 20 min of the formal experiment, which included three simple, three medium, and three difficult questions from the 6–8-year-old categories of the 2018

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Table 1 Demographics of the study population

	Frequency	Proportion (%)
Sex		
Male	16	57.1
Female	12	42.9
Age		
6-Year-old	4	14.3
7-Year-old	16	57.1
8-Year-old	8	28.6
Grade		
First grade	20	71.4
Second grade	8	28.6

and 2019 International Bebras competitions. Each participant had a teaching assistant to help them read the questions. Descriptive statistics showed that the average CT score was 5.93 (SD = 1.92). Additionally, the Shapiro-Wilk test results (p = 0.118) demonstrated that participants' CT scores fitted a normal distribution.

Procedure

At the beginning of the experiment, children participated in a 20-min training activity that helped them to recognize and understand the basic use of the unplugged toolkit. In the next 5 min, each child was accompanied by a teaching assistant to fill out a survey of their perceptions about the unplugged toolkit and activities. Considering that children of this age have immature reading skills, the teaching assistant read each item for them and explained when necessary. In the following phase, a teacher guided children to engage in four unplugged activities within 60 min. Finally, children spent 5 min filling in the survey of their perceptions about the unplugged toolkit and activities again, also with teaching assistants.

Learning Activities

This study followed guidelines of embodied design in materials, activities, and facilitation, which were proposed by Abrahamson and Lindgren (2014). Five main guidelines were adopted (see Table 2). These guidelines are unpacked and used to design the unplugged toolkit and activities.



Fig. 1 A girl is using the CodeWay toolkit

Unplugged Toolkit

The CodeWay unplugged toolkit designed in this research is composed of a series of building blocks with specific functions, as well as large and small balls. As Fig. 1 shows, learners need to build a pathway using CodeWay blocks with different functions and make the ball roll from the Input block to the Output block by way of gravity to complete the task. The process of each ball rolling is like the process of code running, and the process of building a pathway step by step is akin to the process of designing algorithms and solving problems. According to guideline (b), this process mobilizes learners' perceptual senses and kinesthetic coordination. Furthermore, these CodeWay blocks represent particular computational concepts, such as random, sequence, and conditions. This toolkit provides students with different feedback when they use different blocks, establishing the action-environment couplings (guideline (c)).

Table 2 The guidelines of embodied design

	Guidelines
(a)	Activities should draw on students' preexisting ability to mobilize in real or virtual three-dimensional space.
(b)	Students should be provided opportunities to use their perceptual senses and kinesthetic coordination to engage in perception-action loops.
(c)	The learning environment should be conducive to coupling the somatic actions with the environment via action-feedback loops.
(d)	Breakdowns of the action-environment couplings should be gradually introduced by presenting objectives that cannot be met using solutions that the learner has already mastered.
(e)	Real-time physical feedback should be implemented to reinforce these metaphors for knowledge.

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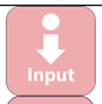
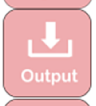
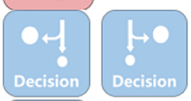
CT concepts	Blocks	Perception mapping	Action mapping
sequence	input		The ball enters from above and falls vertically, realizing the input
	output		The ball falls in from above and stops, realizing the output
	right		The ball falls out from the right side
	left		The ball falls out from the left side
	straight		The ball enters from above and falls out from below
conditions	decision		The large ball falls out from the side; the small ball falls out from below
	toggle		If it is the first input, the ball will fall out from the left; if it is the second input, the ball will fall out from the right, and so on
loop	circle		Bouncing the ball upward to a certain position, thus forming a cycle
random	random		The ball may fall out from the left or the right
and	and		The ball can continue to fall only when two balls fall in (the program can continue only when both conditions are met)

Fig. 2 The mapping relationship between CodeWay blocks and computational concepts

Children should use these blocks to build paths and solve problems in specific situations that are designed to cultivate learners' computational practice (e.g., abstraction, evaluation).

Considering guideline (e), this toolkit was designed to provide students with real-time feedback on the specific movement of balls. These pieces of feedback acted as metaphors for the corresponding computational concepts. Figure 2 presents the mapping relationship between the toolkit and computational concepts. The mapping of these unplugged blocks and computational concepts has two aspects: perception and action. Perception mapping mainly includes visual perception, which is based on the properties of appearance, color, text, icon, and shape. The action mapping contains the movements of balls, such as turning and falling. For example, to complete a condition concept, the students should first use color and icon as the visual senses in perception, then integrate these perception concepts into the specific movements of big balls and small balls. Doing this, the big ball rolls out from the side, and the small ball rolls out from below (Fig. 3 shows the internal structure of the Decision Block).

Unplugged Activities

In accordance with guideline (a), to mobilize students' kinesthetic experience, all activity scenarios are designed to involve the movement of objects in the real 3D world. In the training phase, children participated in an activity called "The Adventure of Big Ball and Small Ball," which included five sub-activities. According to guideline (d), each activity involves learning of a new block. In these activities, children needed to explore the function of each block by themselves and select the appropriate block to handle various simple challenges. After each sub-activity was completed, the teacher explained the corresponding computational concepts in conjunction with real life examples. In the 60-minute learning activity, children completed four tasks (see Table 3).

Data Collection and Analysis

This study used a mixed method to answer the research questions. Quantitative data regarding young children's perceptions about the CT unplugged activities was used to answer RQ1. A survey based on the four-dimensional

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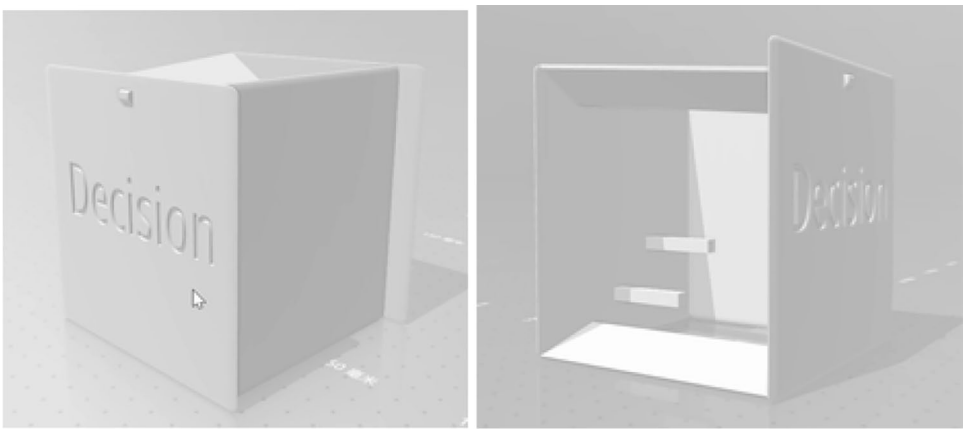


Fig. 3 The structure of decision block

ARCS model—including attention, relevance, confidence, and satisfaction—was conducted to collect the quantitative data. This model was originally proposed to help teachers develop instructional materials with a strong motivational appeal to students (Keller, 1987). In this study, a validated, highly reliable (Cronbach $\alpha = 0.92$) reduction of Loorbach et al. (2015) was used (see Table 4). The survey used a Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree). Furthermore, Wilcoxon tests were used to examine differences in children's perceptions of CT unplugged activities in two surveys after the training phase and after all learning activities had ended.

Video recordings of participants' learning processes were used as qualitative data to answer RQ2 and RQ3. The qualitative data were coded and transcribed to reveal young children's experiences in unplugged activities. Since the videos of two students were incomplete, a total of 26 students' videos were used for analysis. In this study, the computational concepts are mainly embodied by blocks and balls, so when studying the learning process of students, their computational practice was focused. Additionally, a well-cited CT framework proposed by Selby and Woollard (2013) was adopted. Then, based on the CT category and participants' embodied interactions recorded in the videos—defined as the social, verbal, and non-verbal aspects of human communication that occur as people interact with each other and the environment to solve problems (Goodwin, 2007; Streeck et al., 2011)—the corresponding embodied phenomenon under each category was defined (see Table 5). The coding scheme was spirally developed from the qualitative data according to Boeije (2013).

Two researchers conducted multiple rounds of coding on these videos. The first round of coding isolated critical events (i.e., portions of the data that hold the specific elements of investigation) (Powell et al., 2003). According to the coding scheme, the researchers selected critical events ($n = 233$) that captured children's CT, emerging as embodied phenomena. In the second round of coding, the researchers determined which dimension of computational thinking emerged as an embodied phenomenon in each critical event. In the final round of coding, these critical events were transcribed using the multimodal transcript designed by Kopcha et al. (2021). This transcript was designed based on Bezemer's (2014) framework for multimodal analysis and a social semiotic approach to analyze embodied interactions with technology. In this case, the transcript was used to analyze young children's embodied interactions with the unplugged toolkit as they learned CT. Furthermore, this study performed descriptive statistics on the frequency of children's various dimensions of CT that emerged as embodied phenomena.

Results

Do the Students' Perceptions of the Unplugged Activities Improve After Engaging in Them?

From the average score of the first survey in Table 6, it is evident that participants had positive perceptions about the unplugged activities after the training phase. Moreover, according to the Wilcoxon tests, the average score of the second survey is significantly higher than that of the first survey ($z = -3.75$, $p = 0.00$), demonstrating that these activities have a strong motivational appeal to young children. From the scores of the four sub-dimensions, children generally agree that these activities can maintain their attention, make them feel the connection between CT and life, make them confident in learning CT and make them want to continue learning CT. Furthermore, the scores of attention, relevance, and satisfaction significantly improved ($z = -2.79$, $p = 0.00$; $z = -2.69$, $p = 0.00$; $z = -2.02$, $p = 0.04$), indicating that after participating in more complex and longer activities in the formal learning phase, children's perceptions on the unplugged activities become more positive.

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What's the Frequency of Each Students' CT Emerging During Unplugged Activities?

According to the observation and video analyses, various dimensions of CT emerged as embodied phenomena at different frequencies in unplugged activities. Specifically, young children's evaluation (debugging) emerged most frequently (n = 90), followed by algorithmic thinking (n = 57) and generalization (n = 43), while decomposition (n = 20), evaluation (optimization) (n = 20) and abstraction (n = 4) emerged less frequently. This demonstrates that young

Table 3 The content of activities

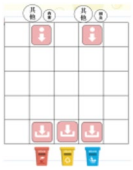
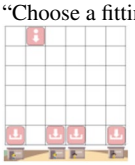
Name	Content	Tools
"Go to the mall" 	Coco and Dede found road construction on the way to the mall. Please help them bypass the obstacles and choose a random road to the supermarket. In this task, the big and small balls represent Coco and Dede respectively, the Input block represents the home, and the Output block represents the shop.	Balls, basic blocks, and Random
"Garbage classification" 	Coco and Dede found garbage on the way to the shop, please help them sort the garbage into the garbage can. In this task, the big and small balls represent different types of garbage, the Input block represents the starting point of throwing garbage, and the Output block represents the garbage can.	Balls, basic blocks, and Decision
"Choose a fitting room" 	After Coco and Dede arrive at the shop, they want to go to the fitting room to try on clothes. Please help them randomly select a suitable fitting room. In this task, the small and big balls represent Coco and Dede respectively, the Input block represents the entrance of the shop, and the Output block represents the fitting room.	Balls, basic blocks, Decision, and Random
"Game challenge"	Each student designs the task by themselves and evaluates how many stars the task is worth. If the opponent does this problem correctly, the opponent will get stars, but if the opponent does it incorrectly, the questioner will get stars.	Free choice

Table 4 Survey about perceptions of CT unplugged activities

Item	Dimension	Description
1	Attention	The content of the activities helps me to maintain attention.
2	Attention	The variety of activities helps me to stay focused in the class.
3	Attention	The way of teaching CT using CodeWay helps me to maintain attention.
4	Relevance	For me, it is clear how these activities are related to my life.
5	Relevance	These activities convey the impression that it is worth learning about CT.
6	Relevance	The CT involved in these activities is useful for me.
7	Confidence	When I study in class, I am sure that I will learn the contents.
8	Confidence	After these activities, I feel confident about passing the CT test.
9	Confidence	The good organization of these activities helps me to be sure that I am going to learn the contents.
10	Satisfaction	I enjoy these activities so much in class, that I would like to learn more about CT.
11	Satisfaction	I like these activities.
12	Satisfaction	It is a pleasure to participate in these well-designed activities.

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Table 5 Coding scheme

CT	Definition	Embodied phenomenon	Example
Abstraction	Construct the representation of problems in symbolic or verbal form	Students describe balls or blocks as objects in the task scenario	[S02-A01] When the small ball arrived at the wrong Output block, a participant said, "Oh no, the little girl went to the men's fitting room."
Decomposition	Break down complex problems into small ones and solve them one by one	When building the path, students first successfully achieve one sub-goal before achieving another sub-goal	[S01-D02] After building the path so that the small ball can successfully reach the goal, a participant tried to build the path to make the big ball reach the goal.
Algorithmic thinking	Design a precise path toward a solution through a set of instructions or rules	According to the task goal, students build the path sequentially	[S03-T01] To accomplish the task 1 goal, a participant placed the blocks one by one.
Evaluation	Ensure an algorithmic solution is a good fit	Debugging: before the task is completed, students find the problem in the path and fix it Optimization: After the task is completed, students make the path more robust or remove redundant blocks	[S05-ED01] A participant placed the ball, locating a problem with the path, and then modified it. [S05-EO01] After completing the task, a participant added the basic path block to avoid occasional situations where the ball might pop out of the path.
Generalization	Apply parts of solutions that have been used in previous situations to new situations quickly	Students use the previously learned blocks correctly and quickly in the new task	[S06-G01] Without the help of the teacher, a participant accurately and quickly selected the Decision block and put it in the correct position.

children engaged the CT thinking process of debugging considerably during unplugged activities.

Furthermore, the box plot in Fig. 4 shows that most students' algorithmic thinking emerged 2–3 times, evaluation (debugging) emerged 2–5 times, and generalization emerged nearly two times, while their decomposition, evaluation (optimization) emerged less than two times. This finding might illustrate that the unplugged activities effectively provided young children with an environment where they could smoothly participate in the learning of CT, especially in evaluation (debugging), algorithmic thinking, and generalization.

How Do Young Children Experience CT as an Embodied Phenomenon During Unplugged Activities?

Various findings regarding how young children experience CT as embodied phenomenon were observed. First, researchers found that most children's (n = 18) CT extended to their bodies and unplugged tools. During the activities, young children used their bodies and resources in the environment to solve problems. For example, as Table 7 shows, the girl used gestures to simulate the falling path of the ball; this was essential for her to design the rest of the path. Similar gestures emerged not only with algorithmic thinking but also with evaluation.

Another typical example illustrates the extension of young children's CT. As Table 8 shows, researchers observed that when designing their solution step by step, many children faced the whiteboard and kept rotating the blocks. They likely did this because when the ball is falling, its left side is actually the students' right side, and it is difficult for children to mentally rotate the icon on the block at first glance. In other words, they used their hands to rotate the block instead of mentally rotating the block. This finding might indicate young children's bodily experiences play an important role in their engagement in algorithmic thinking during unplugged activities.

Furthermore, to understand why a small group of children (n = 8) never used gestures nor rotated the blocks, their prior CT level and performance in the activities were investigated. Interestingly, these children either had an extremely high CT level (8–9 points) or an extremely low CT level (3–4 points). Additionally, their performance greatly differed. The several outstanding children (n = 4) were able to accurately apply previously learned building blocks to new tasks, systematically decompose the tasks, rarely debug, and solve problems quickly. In contrast, the poorly performing children (n = 4) often found it difficult to understand the goal of the task, showed poor generalization ability, and often got stuck by the problem or kept throwing the ball and revising the plan in the wrong direction.

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Table 6 Perceptions about these unplugged activities

	N	First survey		Second survey			p
		Mean	SD	Mean	SD	z	
Attention	28	13.36	1.73	14.11	1.34	− 2.79	0.00**
Relevance	28	13.50	1.77	14.21	1.17	− 2.69	0.00**
Confidence	28	13.89	1.66	14.18	1.61	− 1.18	0.24
Satisfaction	28	13.93	1.51	14.14	2.37	− 2.02	0.04*
Total	28	54.68	5.54	56.64	5.64	− 3.75	0.00**

*$p < .05$, **$p < .01$

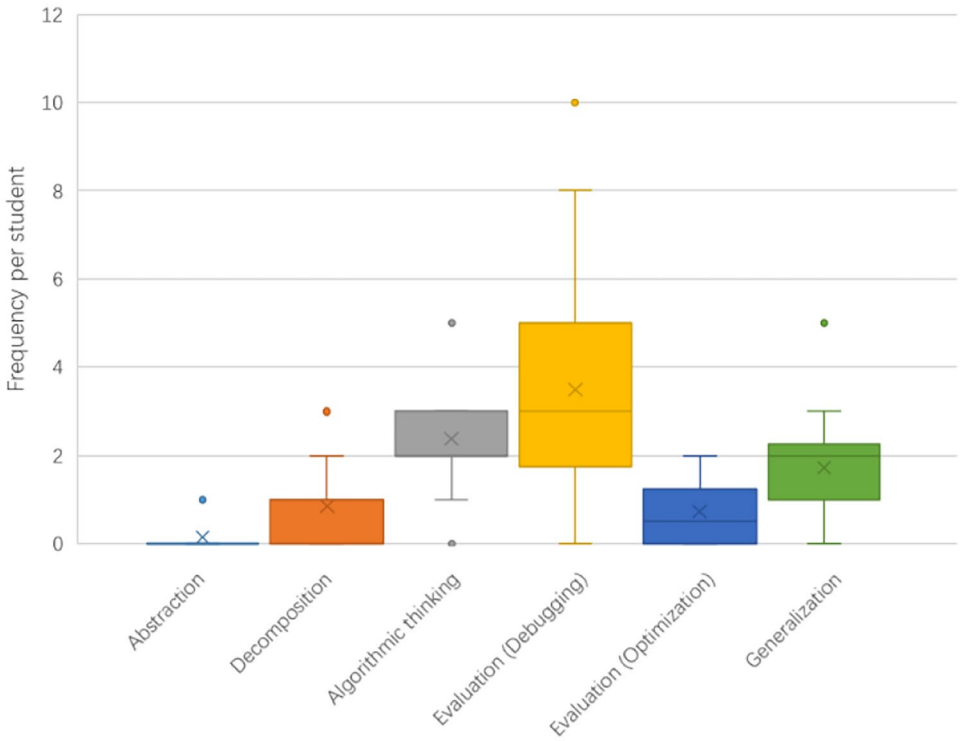


Fig. 4 Frequency of each student's CT emerging as embodied phenomena

Table 7 Using gestures to simulate the falling path of the ball

Embodied Interaction	Transduction
 [She points out the trajectory of the ball with her finger.]	[→ Algorithmic thinking] After building part of the path, she thinks about what to do next by moving her fingers on the whiteboard to simulate the falling path of a big ball. Then she places blocks on the squares where her fingers passed in order.

Researchers also found that CT emerged in perception-action loops. Specifically, some children spontaneously developed some CT such as debugging, decomposing, and generalization during their continuous interactions with the unplugged toolkit without the teacher's instructions. Especially for debugging, which had the highest frequency

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Table 8 Using rotating action to confirm the correct direction

Embodied Interaction	Transduction
 [She rotates the Turn block.]	[→ Algorithmic thinking] When building the path, she uses the Turn block. However, she is not sure whether the turn-right block she chose is correct, so she keeps rotating this block until she is sure that it can make the ball move in the desired direction.

Table 9 Debugging emerged as an embodied phenomenon

Embodied Interaction	Transduction
 [He puts the ball in twice in a row and then places a Turning block.]	[→Evaluation (debugging)] He builds a path thinking that the current plan could make the ball enter from the Input block, bypass the obstacle, and finally reach the Output block. Then he puts in the ball to verify his plan and finds that the ball does not reach the Output Block. He puts in the ball again to observe where the movement of the ball goes wrong. Through this debugging, he successfully locates the problem. There should be a Turning block after the Random block to allow the ball to fall smoothly to the Output block. Afterwards, he immediately adds a Turning block to this position.

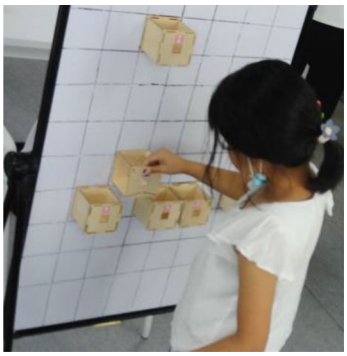
of all CT, the teacher did not teach students how to debug but found that many students spontaneously engaged in debugging. For example, at the beginning of task 1, stu26 was focusing on building the path. At one point, he started to put the ball in the Input Block and found that the ball could not reach the Output Block. Therefore, he began to debug continuously. Table 9 records his first successful debugging. After this successful debugging, he happily

Table 10 Decomposition emerged as an embodied phenomenon

Embodied Interaction	Transduction
 [He first builds a path to ensure the big ball can complete the goal successfully.]	[→Decomposition] After understanding the task, he quickly divides the task into two parts. First, he builds a path so that the big ball can reach the goal smoothly. He then tells the teacher, "I have already built a road, and then I need to build another one." He now considers how to build a path so that the small ball can reach the goal.

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Table 11 Generalization emerged as an embodied phenomenon

Embodied Interaction	Transduction
 [She quickly and accurately places a Random block.]	[→Generalization] The key to this task is correctly using the Random blocks and Decision blocks. After the teacher introduces this task, she immediately says, "I see, just like the last task, I also need to use the Random block here." She keenly captures the computational concept of random and places a Random block, which shows that she has successfully applied her experience using random blocks to solve previous problems to this new task.

told the teacher, "It's better to have a 'fence' so the ball doesn't fly away, which is very good! I didn't add this block for the first time, and the ball kept flying out. Then after I added it, the ball couldn't fly out." In subsequent activities, he was more active in debugging, and the frequency of his successful debugging increased. Similarly, some children demonstrated decomposition (see Table 10) and generalization (see Table 11) skills. It may be inferred from these findings that this study created an effective learning environment that allowed young children to naturally develop their CT through a continuous cycle of perceiving and controlling objects, as guideline (c) stipulates.

Discussion

The results of the survey regarding children's perceptions show that these activities have a strong motivational appeal to children. Although several researchers have introduced embodied cognition perspectives into the teaching of CT for young children (Clarke-Midura et al., 2021; Saxena et al., 2020; Sung et al., 2017), they have usually focused on students' cognitive processes or results and ignored their subjective feelings. The findings in this study illustrate that young children themselves agree that these activities could indeed attract them. At the same time, researchers have suggested that more research is needed to study young children's motivations for learning CT (del Olmo-Muñoz et al., 2020).

According to the video analyses, algorithmic thinking, evaluation (debugging), and generalization occur frequently. In previous activities designed to help young children learn CT, young children's algorithmic thinking often emerged (Chen & Chi, 2020; Saxena et al., 2020), but young children rarely showed their thinking process of debugging and generalization. A possible explanation for this might be that the toolkit and activities designed based on the guidelines of embodied design in this study are more advantageous. Specifically, unlike common educational robots like bee-bot (Saxena et al., 2020; Angeli & Valanides, 2020), the CodeWay unplugged toolkit provides a visual representation of the commands (i.e., blocks) to reduce the cognitive load on young children's limited memory resources. As bee-bot does not have this feature, students usually forget the sequence of commands they have entered (Angeli & Valanides, 2020). And, compared to board games (Chen & Chi, 2020), our unplugged activities give real-time physical feedback via the movement of balls, which allows young children to easily and naturally engage in the thinking process of CT. These findings provide insight into how tools and activities should be designed for young children; it is important to provide them with visual representation and real-time physical feedback.

Furthermore, the analysis of qualitative data revealed how young children's CT emerges as an embodied phenomenon. First, an encouraging finding shows that CT extended to children's bodies and unplugged toolkits. Kopcha et al. (2021) had similar findings after analyzing children's embodied interactions during educational robot activities. They found participants would simulate the robot's actions and convert them into a computer program. Participants in this study were also observed using gestures to simulate the falling path of the ball, illustrating that the critical learning tools (e.g., robots, balls) became the objects that helped children connect CT with their own bodily experience (Papert, 1980). Similarly, when designing the route, children often kept physically rotating the Turning block to determine whether the block was correct, effectively reducing the load of mentally rotating the image. This finding may help us better understand the positive correlation between CT and mental rotation ability demonstrated by Città et al. (2019). When learners used CT during these activities, they needed to use their physical experience in relation to space (Manches et al.,

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2020). Furthermore, teachers should recognize that young children may find mental rotation challenging when solving CT problems and give timely guidance. For example, teachers may guide students to use their own bodies or resources around them to rotate images or identify directions.

Moreover, while past researchers paid greater attention to the specific embodied interactions of students (Clarke-Midura et al., 2021; Kopcha et al., 2021; Sung et al., 2017), this study combined the analysis of the computational thinking level of the students. An interesting finding showed that there was a small group of students with either extremely high or low CT levels who never used gestures or rotating actions during the unplugged activities. This result might suggest that gestures act as a scaffolding to help children learn CT, and when children have rather high-level CT, their gestures fade away. This suggestion is underpinned by opinions of Goldin-Meadow (2015), who argued that the use of particular gestures could support learning. The educational implications of this finding might be that teachers should encourage students to use gestures that could make implicit knowledge explicit (Broaders et al., 2007).

The second major finding regarding the way children's CT emerged involved the appearance of CT in perception-action loops. Previous researchers have noted that there are bidirectional, reciprocal relations between perception and action (Gibson, 2014). The emergence of CT in perception-action loops may be explained by the fact that during the process of constant perception and actions, students gradually develop new perceptuomotor schemas that enable them to effectively control objects (Abrahamson & Lindgren, 2014). In this case, the new perceptuomotor schemas may refer to certain aspects of CT. Specifically, as children constantly perceive the movement of the ball and building blocks, they may gradually develop a new perceptuomotor schema to manipulate the blocks and the ball, such as putting in a ball to determine the problem in the path of the blocks—a process of debugging. This finding inspired us when teaching CT to young children, it is critical to design couplings between the knowledge concepts and somatic actions and create a learning environment that make young children's body engagement which optimally facilitate conceptual development occur naturally.

Conclusion

This study introduced the embodied cognition perspective into CT education, innovatively applying embodied design principles to the creation of unplugged tools and activities for young children. Results indicate that children had positive perceptions of the unplugged toolkit and activities. Additionally, the unplugged toolkit positively impacted children's learning of CT. Furthermore, based on the video analyses, this study observed CT to extend to the unplugged toolkit and children's bodies. CT also emerged in perception-action loops, offering insight into how children's CT emerges as embodied phenomena during unplugged activities. Although the sample size of this study was small and the length of instruction was relatively short, this study presents a unique contribution to helping educators understand children's process of learning CT more deeply from an embodied cognition perspective and revealing the connection between their body-based actions and CT, which provides enlightenment for the instructional design of young children's CT.

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Declarations

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